

Siqi Xiao (She/Her)

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Education

University of Southern California, MS in Computer Science - Applied Data Science Aug 2023 - May 2025

- **Coursework:** Machine Learning for Data Science, Applied Natural Language Processing, Data Management, Applications of Data Mining, Data Visualization

Shanghai Jiao Tong University, BS in Electrical and Computer Engineering Sept 2019 – Aug 2023

- Minors: Data Science, Entrepreneurship
- Honors: 1st class Outstanding Scholarship, Academic Progress Scholarship
- **Coursework:** Data Structures and Algorithms, Introduction to AI

Skills

Programming Languages: Python/PySpark, SQL, C, C++

Frameworks & Technologies: PyTorch, TensorFlow, Sklearn, MongoDB, Firebase, Hadoop, Apache Spark, Tableau, PowerBI, AWS, GCP, Linux, Git

Skills: Machine Learning (Regression, Trees, Clustering, etc), Modeling (LSTM, Transformer, Neural Network, etc), Data Visualization, A/B Testing, Statistical Analysis, Probabilistic in Engineering, Cross-functional collaboration, FMVA (ongoing)

Experience

Machine Learning Engineer Intern, 211 LA – Los Angeles, CA June 2024 – Aug 2024

- Developed a chatbot powered by Retrieval-Augmented Generation (RAG) for the website, enhancing response accuracy and improving call handling efficiency by 32% and user satisfaction rates by 11%.
- Analyzed five years of caller data using Tableau to create detailed profiles, identifying key demographic segments and frequently asked questions to inform retrieval corpus construction.
- Conducted internal testing of the RAG pipeline, iteratively improving performance by expanding taxonomy, refining retrieval protocols, and integrating real call transcripts, leading to an expected 108% improvement in first-time correspondence and referral success rates.

Projects

Caption of My Day: A Model for Multi-Image Captioning using NLP Techniques Feb 2025 – May 2025

- Led and implemented an advanced image captioning model utilizing encoder-decoder architecture with attention mechanisms and beam search optimization.
- Implemented a custom summarization module using a fine-tuned T5 large language model to generate coherent paragraph-style descriptions from multiple image captions.
- Built scalable data pipelines to process and transform large-scale datasets (e.g., COCO), streamlining model training and evaluation workflows.

SparkYelp: A Hybrid Recommendation System Jan 2025 – May 2025

- Designed and implemented a hybrid recommendation system using Apache Spark and Python combining user-based collaborative filtering with XGBoost regression to predict user ratings on a large scale dataset, achieving an 11.2% reduction in RMSE compared to base model.
- XGBoost Regression model: Integrated complex feature sets into the XGBoost model, leveraging both statistical and NLP-derived insights; Performed cross-validation to fine-tune hyperparameters and enhance model accuracy.
- Collaborative Filtering: Built efficient Spark-based data pipelines, reducing data processing time by 46% compared to Python-only implementations.